Analyzing Gaming Engagement Patterns: A Statistical Framework for Evaluating Genre Preferences and Player Behavior

King Dheniel N. Concepcion[1,*]

[1]College of Computing and Information Technologies, National University - Manila
*concepcionkn@students.national-u.edu.ph

Abstract. This study examines digital gaming engagement and player behavior patterns through a quantitative statistical framework, utilizing Descriptive Statistics and Spearman's Rank Correlation to analyze a personal digital library of 72 active games. By transforming raw historical logs from 2021 to 2026 into structured datasets, the research identifies core drivers of player commitment and interest shifts. Through a rigorous comparative analysis of engagement metrics, the Simulation and Action genres were identified as the historically dominant categories, with Simulation alone accounting for 824.2 hours of total playtime. However, temporal analysis reveals a recent tactical shift toward the Shooter genre, which led engagement in the final 30 days of the study period. The framework produced a strong Spearman Correlation of 0.7942, suggesting a precise association between session frequency and total duration, alongside a highly right-skewed distribution and a statistically significant p-value of <0.001. These findings confirm that gaming behavior follows a predictable power-law pattern where a select few titles, such as Jurassic World Evolution 2 and Grand Theft Auto V, dominate total Player attention. Ultimately, this study provides a reliable methodology for analysts to leverage personal archives for objective behavior modeling and future media consumption prediction.

Keywords: Digital Gaming Library, Descriptive Statistics, Spearman's Rank Correlation, Player Engagement, Playnite, Data Visualization, Quantitative Analysis.

1 Introduction

In the modern digital age, tracking video game engagement has become a vital method for understanding personal digital consumption habits. By utilizing universal library managers like Playnite to aggregate data from platforms such as PC and consoles, this study, titled "Statistical Analysis of Digital Gaming Library," establishes a systematic framework to transform raw playtime metrics from 79 games into structured insights [1]. Despite the abundance of such data, personal records often remain unexamined, making it difficult to identify core engagement drivers within highly skewed environments. This research addresses this gap by applying a quantitative statistical framework to demonstrate that digital consumption follows predictable, evidence-based patterns, moving beyond raw logs to reveal true player preferences over multiple years of activity [2]. This systematic approach ensures that personal gaming history is no longer just a list of titles, but a verifiable dataset that reflects behavioral trends. By establishing this groundwork, the study highlights how individual data can be leveraged to understand the broader relationship between technology and player commitment.

The primary objective of this study is to utilize mathematical modeling and non-parametric tests, specifically Spearman's Rank Correlation (ρ), to explain gaming behavior according to professional scientific standards. The research seeks to characterize the library's distribution profile to identify power-law patterns and high-engagement "outlier" titles while evaluating the relationship between session frequency and total play duration [3]. Furthermore, it analyzes the temporal evolution of preferences across different platforms and categories to identify shifts in player interest over time. Ultimately, this framework provides a professional guide for researchers to analyze personal data, demonstrating that even a personal hobby can be studied deeply to uncover meaningful facts about technology use in a resource-constrained digital environment [4]. These findings offer a clear methodology for students and analysts to apply rigorous statistical scrutiny to their own digital archives. Through this analysis, the study confirms that consistent engagement metrics can provide a reliable roadmap for predicting future media consumption and interest shifts.

2 Review of Related Literature

The growth of digital gaming has made it necessary to find better ways to track and understand how people interact with their software collections. As digital libraries grow larger, researchers are moving away from simple observations and are now using data science to study player engagement [5]. This chapter looks at how library management tools help collect data, identifies the need for better statistical methods to analyze playtime, and explains why testing hypotheses is important for understanding modern gaming habits.

2.1 Evolution of Game Management: From Platforms to Universal Libraries

The shift from physical discs to digital downloads has changed how gamers manage their collections. Early studies focused on how digital platforms like Steam, Epic Games, or Ubisoft store data, but these platforms often keep information separate from each other. The rise of universal managers like Playnite has changed this by allowing players to combine data from many different sources into one organized place [6]. These tools are essential because they provide raw information such as "play count" and "time played," which are key metrics for measuring how much a person likes a game [7]. While earlier research mostly looked at "digital hoarding" or how many games a person owned, current studies focus more on the quality of time spent. This means instead of just looking at the number of games, researchers now use data cleaning to make sure the playtime numbers are accurate. For example, removing games that were opened but never played helps ensure that the statistical analysis truly reflects the player’sbehavior. Using these tools allows for a much more detailed look at how a person balances their time across many different titles.

2.2 Statistical Analysis of Engagement and Research Gaps

There is a clear gap in research when it comes to analyzing personal, small-scale gaming libraries using formal statistical tests. Many studies focus on "Big Data" from millions of anonymous players, often ignoring the unique habits and personal choices of individual players [8]. Most current literature suggests that playtime is heavily influenced by the game genre, but there is not enough proof showing how different platforms, like PC versus handheld consoles, change these habits in a personal setting [9]. By using tests like Spearman’s Rank Correlation (ρ) and detailed correlation analysis, researchers can prove if a person’s favorite genre matches where they spend the most time or if they are just playing certain games because they are easily accessible. Correlation analysis helps determine if the number of times a game is opened (play count) leads to more total hours played [10]. This study fills that gap by using a personal dataset of 79 games to show how simple statistical tools can reveal deep insights into digital behavior. This approach moves beyond simple guessing and uses actual data to explain how a modern gaming library is used.

3 Methodology

This section, the systematic process of data collection, preparation, and statistical analysis is described. The study follows a quantitative approach to analyze personal gaming habits using a localized dataset.

3.1 Participants

The subject of this research is a 22-year-old fourth-year student currently pursuing a Bachelor of Science in Computer Science at National University - Manila. Given the subject’s academic specialization in computer science, there is an inherent technical proficiency in managing digital ecosystems. Beyond academic requirements, the subject is identified as a dedicated gaming enthusiast and a collector of action figures. This background provides the necessary context for the maintenance of a diverse and extensive digital library, which serves as the primary data source for this investigation.

3.2 Data Collection Methods

The data was collected using Playnite, an open-source video game library manager that aggregates titles from various services such as Steam, Epic Games, and PlayStation. The dataset covers several years of historical activity, finalized on February 13, 2026.

- Variables Collected: The study utilized all ten (9) attributes extracted from the database to ensure a comprehensive analysis: Name, Categories, Added, Last

Played, Recent Activity, Completion Status, Platforms, Play Count, and Time Played.
- Frequency of Logging: Data logging was performed automatically by the Playnite software, which records the launch frequency and the exact duration of each gaming session without manual intervention.
- Sample Size: The final dataset consisted of 79 unique game titles across various digital ecosystems.

3.3 Operational Definitions

Table 1 Provides a detailed breakdown of the variables utilized in this study, including their specific definitions and the metrics used for measurement. These operational definitions ensure that the automated data extracted from the digital library is standardized for statistical analysis, allowing for a precise evaluation of player engagement, platform distribution, and chronological library growth.

Table 1. Definition of Variables

Variables	Description
Name	The official title of the digital game is used for identification.
Categories	The genre or thematic classification (e.g., RPG, Shooter, Arcade) assigned to the game.
Added	The specific date and time the game was first registered in the digital library.
Last Played	The timestamp recording the most recent date and time the player played the game.
Recent Activity	The most recent interaction recorded by the library manager.
Completion Status	The player-defined progress level (e.g., Played, Plan to Play) describing the game's state in the library.
Platforms	The hardware or software environment used to run the game (e.g., PC, Sony PlayStation 3, Nintendo Switch).
Play Count	The total number of sessions or "launches" recorded for a specific game.
Time Played	The cumulative duration spent actively running a game, normalized from raw seconds into hours (H=S/3600) for statistical interpretation.

3.4 Data Cleaning and Preprocessing

To maintain the integrity of the statistical results, the raw dataset underwent a systematic preprocessing phase as documented in the analytical notebook:

- Unit Normalization: The Time Played variable was transformed from raw seconds to hours to ensure better interpretability. This was achieved by dividing the total seconds by 3,600. This conversion standardized the units across the dataset, allowing for more precise comparisons of long-term engagement.
- Categorical Correction: An audit of the Categories column was conducted to resolve typographical inconsistencies and remove trailing spaces. The researcher standardized the following seven (7) core categories: Fighting, Shooter, Survival, Simulation, Action, Arcade, and RPG. By unifying these labels, the researcher ensured that each game title was correctly grouped, preventing the fragmentation of data during category-based analysis.
- Missing Value Imputation: For titles lacking genre metadata in the source file, an "Uncategorized" label was applied. This approach allowed the researcher to maintain the full sample size of 79 titles without introducing the bias associated with data exclusion.
- Date Normalization: The Added, Last Played, and Recent Activity columns were parsed as datetime values and then converted to date only (time component removed). This step ensured that temporal analysis such as games added per year, playtime per year, and recent activity within the past 30 days used calendar date only and was not affected by time-of-day differences in the source timestamps.
- Status Filtering: The dataset was partitioned to prioritize "Played" games. The analysis focused specifically on titles with a Play Count greater than zero to accurately reflect the subject's active gaming habits rather than the total library acquisition.

3.5 Exploratory Data Analysis (EDA) and Statistical Techniques

This study employed a quantitative descriptive-correlational design using Exploratory Data Analysis (EDA) to identify patterns and relationships within the gaming library. The following statistical models and techniques were implemented:

3.5.1 Descriptive Statistics

To summarize the central tendency and dispersion of the gaming data, the mean mean ($\bar{x}$), median ($\tilde{x}$) and standard deviation (s) were calculated. Additionally, Percentiles (25th and 75th), Skewness, and Kurtosis were analyzed to determine the distribution profile, tail-heaviness, and variability of the subject's gaming habits. These metrics provided the necessary justification for using non-parametric tests.

$$\bar{x} = \frac{\sum x_i}{n} \quad , \quad s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} \tag{1}$$

Eq. (1) These metrics, along with median and skewness values, helped determine the distribution profile and variability of the subject's gaming habits.

3.5.2 Pearson Correlation Coefficient (r)

The Pearson correlation was used to measure the linear relationship between session frequency and total duration.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{2}$$

Eq. (2) This test assessed whether a direct linear trend exists between how often a game is launched and the total accumulated hours.

3.5.3 Spearman's Rank Correlation (ρ)

Since the gaming data exhibited a highly skewed distribution (non-normal), the Spearman rank correlation was utilized as the primary non-parametric measure.

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)} \tag{3}$$

Eq. (3) This coefficient was used to evaluate the strength and direction of the monotonic relationship between Play Count and Time Played Hours, ensuring a robust analysis of non-linear engagement.

3.5.4 Data Visualization

All findings were visualized using Matplotlib and Seaborn. Analytical tools such as histograms for frequency distribution, box plots for category variance, and scatter plots for correlation trends were used to present the results in a professional and replicable format.

The Statistical Techniques used in the analysis are:

1. Descriptive Metrics used: mean ($\bar{x}$), median ($\bar{x}$) and standard deviation (s) to identify the central tendency and the typical engagement level within the dataset.
2. Spearman's Rank Correlation (ρ) used: a robust non-parametric measure to evaluate monotonic relationships, specifically implemented to address the non-normal skewness of playtime and session frequency. Pearson Correlation (r) used a secondary linear metric to identify straight-line associations and verify consistent engagement trends across various game titles.
3. Visualization Framework used: Matplotlib and Seaborn to generate histograms for frequency tracking, box plots for category variance, and scatter plots for regression.
4. Categorical Mapping used: seven standardized genre labels including RPG and Shooter to ensure consistent grouping for comparative analysis.
5. Unit Normalization used: conversion of raw seconds into hours to ensure interpretable and professional metric reporting.

4 Results

This section presents the results using objective metrics, statistical tests, and visualizations to report the data observed patterns and trends. Each figure and table is accompanied by a technical description to ensure the findings are organized and presented in a purely factual manner.

4.1 Overall Dataset Overview

The gaming library dataset consists of 79 total games, with the primary analysis focusing on a subset of 72 titles that hold a "Played" status and a play count greater than zero. These active games span six distinct platforms, including the Nintendo Switch, Nintendo Switch 2, PC (Windows), and Sony PlayStation 3, 4, and 5, while being organized into seven core categories: Action, Arcade, Fighting, RPG, Shooter, Simulation, and Survival. By filtering titles with confirmed session activity, this 72-game dataset serves as the standardized basis for all descriptive and correlational statistical measures presented in this report.

4.2 Descriptive Statistics

Table 2 Presents descriptive statistics for Time Played Hours and Play Count. For Time Played, the mean is 29.20 hours, significantly higher than the median of 3.81 hours, with a standard deviation of 82.96 hours. For Play Count, the mean is 35.42 sessions compared to a median of 6.00, with a standard deviation of 94.07. Both variables exhibit strong right-skewed distributions, as indicated by skewness values of 4.90 and 4.69, respectively. These metrics, alongside the quartiles and extreme kurtosis, confirm that gaming engagement is concentrated in a few highly played titles while most of the library sees minimal activity.

Table 2. Descriptive Statistics Summary

Statistic	Time Played (Hours)
Mean	29.2
Median	3.81
Standard Deviation	82.96
Minimum	0
Maximum	570.58
25th Percentile	1.31
75th Percentile	13.8
Skewness	4.9

Statistic	Time Played (Hours)
Kurtosis	27.44

Figure 1 Displays a histogram visualization showing the distribution of Time Played Hours across all 72 games in the dataset. The histogram uses bins to show the frequency of games at different playtime levels, with the x-axis representing hours played and the y-axis representing the number of games. Mean (29.20 hours) and median (3.81 hours) values are marked with vertically dashed lines to highlight the difference between average and typical values. The histogram clearly shows that most games cluster at low playtime values (below 50 hours), with a long tail extending to the right, indicating that a small number of games have exceptionally high playtime values up to 570.58 hours.

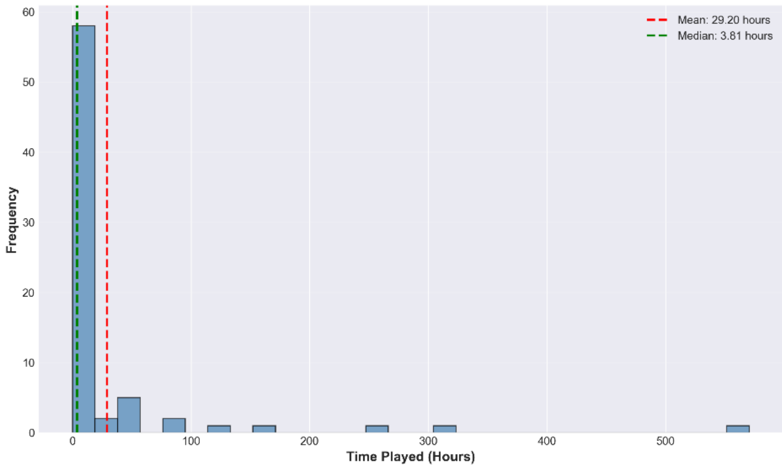

Fig. 1. Histogram of Time Played Hours

Figure 2 Displays a histogram visualization showing the distribution of Play Count (number of play sessions) across all 72 games in the dataset. Like Figure 1, this histogram uses bins to show the frequency of games at different session count levels, with the x-axis representing number of sessions and the y-axis representing the number of games. Mean (35.42 sessions) and median (6.00 sessions) values are marked with vertical dashed lines. The histogram reveals that most games have been played only a few times (typically less than 20 sessions), with a small number of games showing much higher session counts, reaching up to 585 sessions, creating a right-skewed distribution pattern.

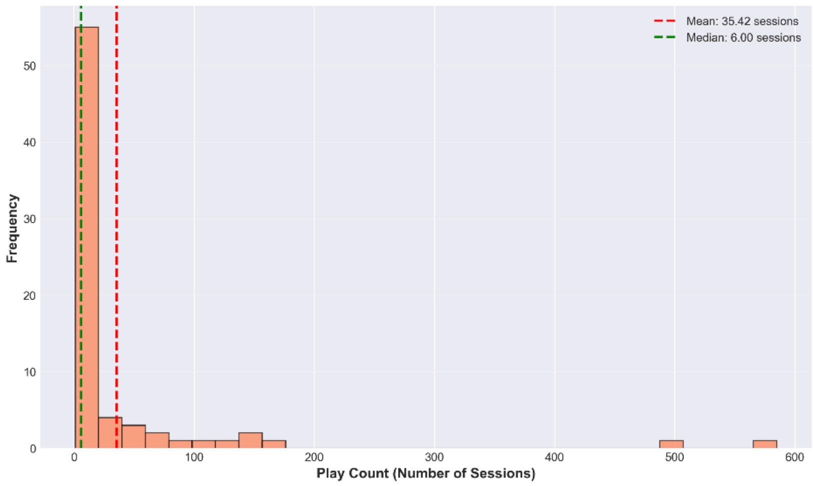

Fig. 2. Histogram of Play Count

4.3 Time-Series Trends

Figure 3 Shows a bar chart displaying the number of games added to the library each year from 2021 to 2026. The x-axis represents the years (2021, 2022, 2023, 2024, 2025, 2026), and the y-axis represents the count of games added in each year. Each bar's height corresponds to the number of games acquired during that specific year. This chart reveals patterns in library growth over time, showing which years had the highest game acquisition rates and identifying trends in how the gaming library expanded across the six-year period.

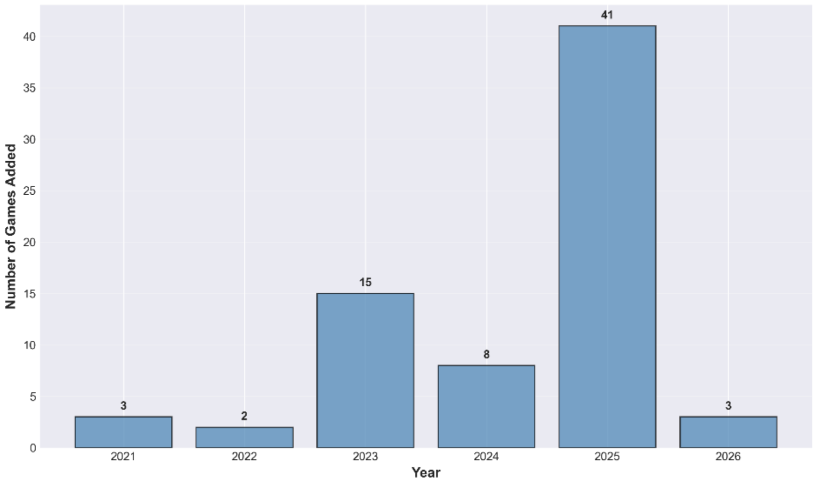

Fig. 3. Games Added Per Year (2021-2026)

Figure 4 Shows a line chart displaying the total playtime (in hours) spent playing games each year from 2021 to 2026, regardless of when the games were added to the library. The x-axis represents the years, and the y-axis represents the total hours played. Each point on the line represents the total playtime for that year, connected by lines to show the trend over time. The line chart reveals fluctuations in gaming activity across different years, showing peak years of high engagement and periods of lower activity, allowing identification of temporal patterns in gaming behavior.

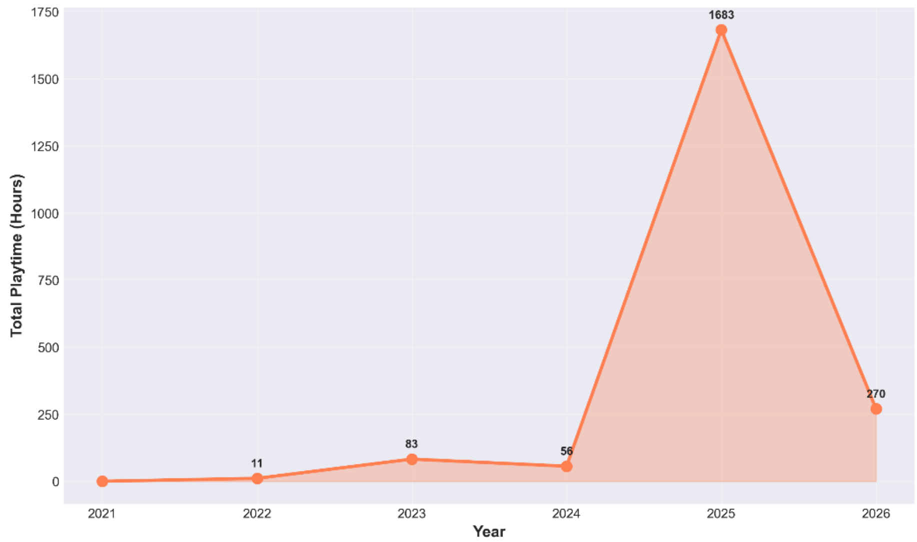

Fig. 4. Total Playtime Per Year (2021-2026)

4.4 Correlation Analysis

Figure 5 Presents a correlation matrix heatmap visualization showing the relationships between numeric variables in the dataset. The heatmap uses a color gradient to display correlation coefficients, with values ranging from -1 (perfect negative correlation, typically shown in blue/cool colors) to +1 (perfect positive correlation, typically shown in red/warm colors). The correlation matrix specifically shows the relationship between Time Played Hours and Play Count, with the correlation value displayed in each cell. The diagonal typically shows perfect correlation (1.0) for each variable with itself. This visualization provides a quick, intuitive overview of the correlation strength between the two-engagement metrics.

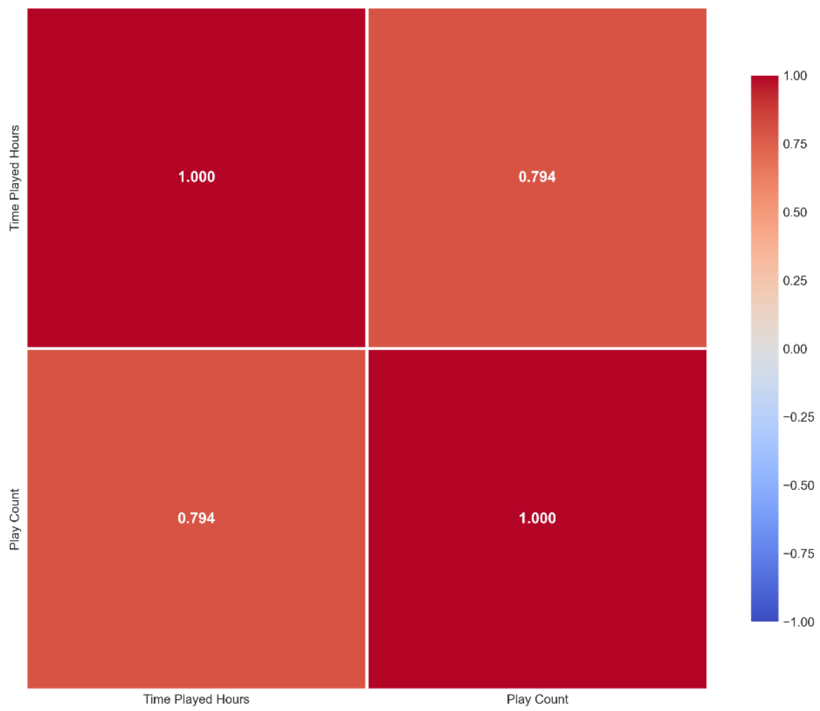

Fig. 5. Correlation Matrix Heatmap

Table 3 Presents the correlation analysis results and statistical test outcomes between Time Played Hours and Play Count. The table reports both Pearson correlation coefficient (measuring linear relationships) and Spearman correlation coefficient (measuring monotonic relationships, more robust to outliers). The table includes the correlation coefficient values, p-value, and significance level from the Spearman correlation test. The Spearman correlation coefficient is 0.7942, indicating a strong positive relationship between the two variables. The table also indicates whether the relationship is statistically significant based on the p-value threshold of 0.05.

Table 3. Summary Comparative Analysis

Variable Pair	**Pearson (r)**	**Spearman (ρ)**	**p-value**
Time Played Hours vs Play Count	0.7942	0.7942	< 0.001

Figure 6 displays a scatter plot visualization showing the relationship between Play Count (x-axis) and Time Played Hours (y-axis). Each point in the plot represents one game from the dataset, positioned according to its Play Count and Time Played Hours values. The scatter plot reveals how these two variables relate to each other, with a

trend line (regression line) overlaid on the plot to indicate the direction and strength of the relationship. Points clustering along an upward-sloping trend indicate that games with higher play counts tend to have higher total playtime. Outliers (games with unusually high values) are visible as points far from the main cluster. The Spearman correlation test shows a correlation coefficient (ρ) of 0.7942. The p-value is less than 0.001, which is below the significance threshold of 0.05. Therefore, the relationship between Play Count and Time Played Hours is statistically significant. The effect size, represented by the correlation coefficient of 0.7942, indicates a strong positive relationship.

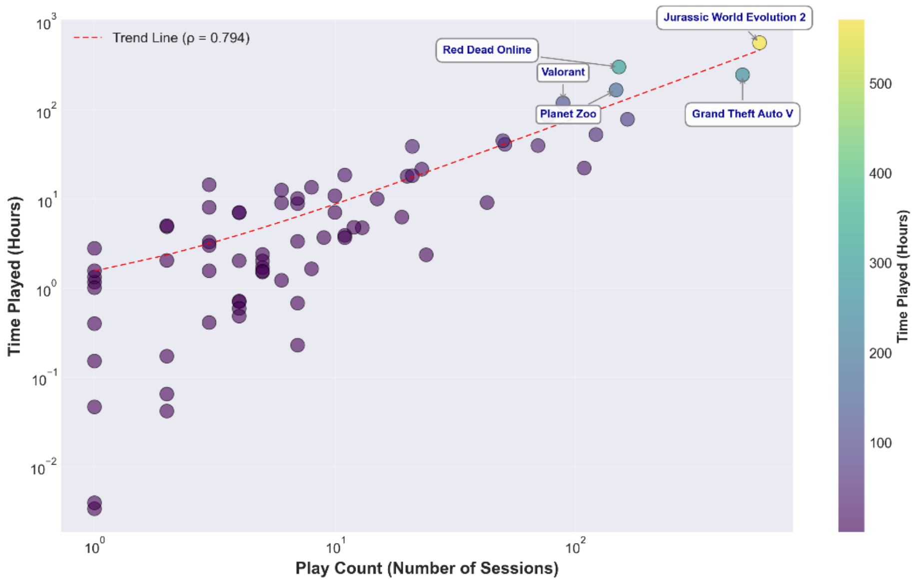

Fig. 6. Scatter Plot of Play Count vs. Time Played Hours

4.5 Platform Distribution Analysis

Figure 7 Shows a box plot visualization displaying the distribution of playtime (Time Played Hours) across different gaming platforms in the dataset. The x-axis represents the different platforms (Nintendo Switch, Nintendo Switch 2, PC (Windows), Sony PlayStation 3, Sony PlayStation 4, and Sony PlayStation 5), and the y-axis represents Time Played Hours. Each box plot shows the median value (middle line), the first quartile (bottom of box), the third quartile (top of box), and whiskers extending to show the range of values. Outliers are displayed as individual points beyond the whiskers. This visualization makes it easy to compare gaming activity across different platforms, identifying which platforms have higher median playtime, greater variability, or more outliers, revealing platform-specific engagement patterns.

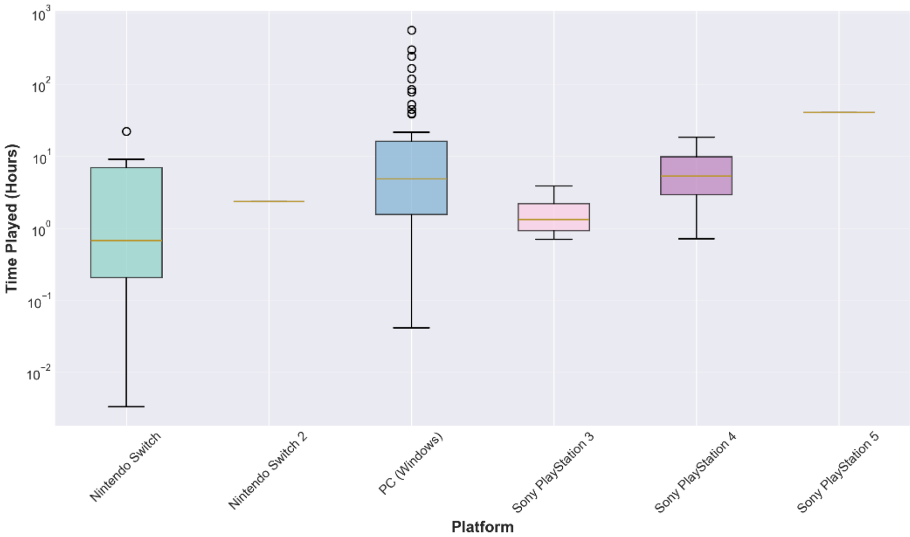

Fig. 7. Scatter Plot of Play Count vs. Time Played Hours

4.6 Category Analysis

Table 4 Presents a comprehensive category analysis summary showing three key metrics for each game category: total playtime (sum of all hours played across all games in that category), number of games (count of games belonging to each category), and average playtime per game (total playtime divided by number of games). The table includes all categories: Action, Arcade, Fighting, RPG, Shooter, Simulation, and Survival. This table identifies which game category receives the most engagement based on total hours played, which category has the most games, and which category has the highest average playtime per game, providing insights into genre preferences and engagement patterns.

Table 4. Category Analysis Summary

Categories	Number of Games	Total Hours Played	Avg Hours per Game	Median Hours	Total Play Sessions	Avg Play Sessions
Simulation	9	824.25	91.58	1.57	912	101.33
Action	16	763.96	47.75	8.22	874	54.62
Shooter	14	231.23	16.52	6.06	204	14.57
Survival	10	159.41	15.94	13.1	151	15.1

Categories	Number of Games	Total Hours Played	Avg Hours per Game	Median Hours	Total Play Sessions	Avg Play Sessions
Arcade	12	79.5	6.63	1.6	230	19.17
RPG	4	24.7	6.17	1.19	135	33.75
Fighting	7	19.56	2.79	2.03	44	6.29

Figure 8 Displays a bar chart showing the top category (genre) per year from 2021 to 2026. The x-axis represents the years, and the y-axis represents the game categories. Each bar shows which category had the highest total playtime in that specific year. The chart reveals how the most-played genre changed from year to year, showing shifts in gaming preferences across different years. This visualization helps identify whether certain genres consistently dominate or if preferences vary significantly over time, revealing temporal patterns in genre engagement.

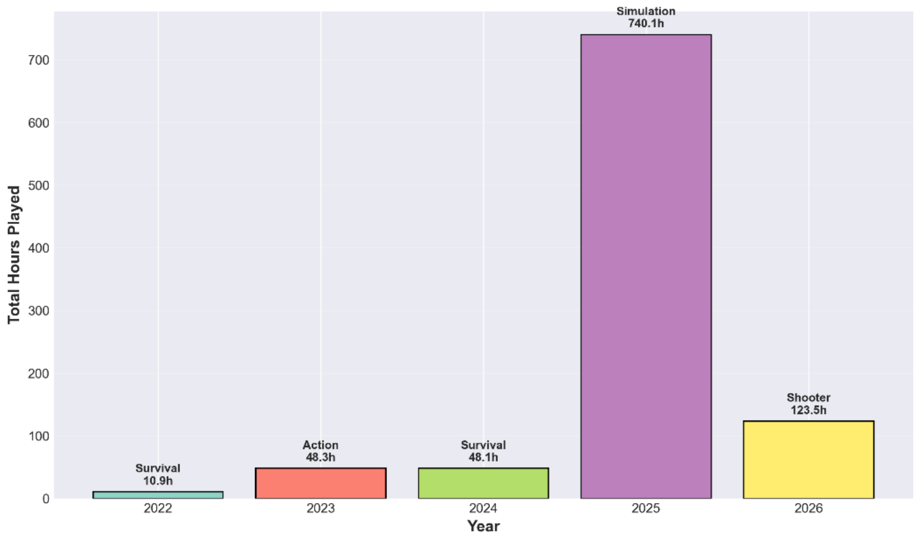

Fig. 8. Top Category Per Year (2021-2026)

Figure 9 Playtime by Category (Past 30 Days) displays a bar chart showing playtime by category for games played in the past 30 days. The x-axis represents the different game categories, and the y-axis represents the total hours played in each category during the recent 30-day period. Each bar's height corresponds to the total playtime for that category. This chart shows recent gaming activity, providing a short-term view of current engagement patterns and identifying which genres are currently receiving the most attention, which may differ from long-term historical patterns shown in other analyses.

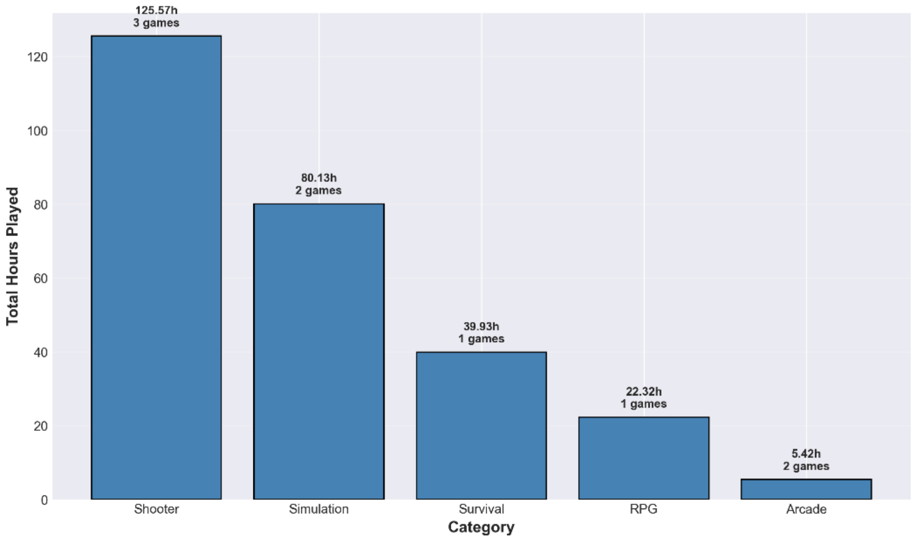

Fig. 9. Playtime by Category (Past 30 Days)

4.7 Summary of Statistical Analysis

Table 5 All key findings from the statistical analysis in one comprehensive reference table. The table includes the sample size (number of games analyzed), descriptive statistics for both Time Played Hours and Play Count (mean, median, and standard deviation), the Spearman correlation coefficient between the two variables and its statistical significance (p-value), and identifies the top categories overall (based on total playtime across all years) and in the past 30 days (based on recent activity). This table serves as a quick reference summary, allowing readers to see all major findings briefly without referring to multiple tables or figures throughout the Results section.

Table 5. Summary of Statistical Analysis

Component	Result
Sample size (played games)	72
Time Played (Hours) - Mean	29.2
Time Played (Hours) - Median	3.81
Standard Deviation (Time Played Hours)	82.96
Play Count - Mean	35.42
Play Count - Median	6

Component	Result
Standard Deviation (Play Count)	94.07
Spearman ρ (Time Played vs Play Count)	0.7942
p-value	< 0.001
Significance (Î± = 0.05)	Significant
Correlation interpretation	Strong positive correlation
Top category (overall by total hours)	Simulation (824.2 h)
Top category (past 30 days)	Shooter (125.6 h)

4.8 Key Visualizations Summary

The findings are presented through nine figures and four tables. The histograms for Time Played Hours (mean 29.20h, median 3.81h) and Play Count (mean 35.42, median 6) reveal a right-skewed engagement pattern, where a few outliers reach up to 570.58h and 585 sessions. Temporal analysis shows library growth peaking in 2025 with 41 games added and engagement peaking that same year at 1,683h. The strong relationship between session frequency and duration ($\rho = 0.7942$, $p < 0.001$) is visualized via a correlation heatmap and a scatter plot with a trend line. Platform-specific variances are illustrated through a box plot of playtime distribution. Finally, genre analysis highlights a shift from Survival (2022) to Simulation (2025) and Shooter (2026), with Simulation (824.2h) and Action (764.0h) remaining the top categories overall. All key findings are consolidated in the final summary table for quick reference.

5 Discussion

This section interprets the findings within the context of gaming behavior and existing literature, highlighting unexpected outcomes and possible reasons for observed patterns. It also addresses research limitations, such as sample size and bias, while offering recommendations to guide future studies and broader knowledge.

5.1 Interpretation of Results

Table 6 The analysis of the dataset reveals distinct behavioral patterns in the participant's gaming habits, as summarized in Table 6. The results indicate a highly selective engagement style, where a few high-engagement titles such as Jurassic World Evolution 2 and Grand Theft Auto V account for the vast majority of total playtime. This concentration suggests that while the participant acquires many games, personal interest is heavily focused on deep-engagement titles within the Simulation and Action genres. Furthermore, the participant's behavior exhibits a strong "session-to-playtime" loop; the high Spearman correlation ($\rho = 0.7942$) proves that as the participant opens a game more frequently, it directly leads to significantly higher total time investment.Temporal trends show that the participant's gaming activity is not constant, with a massive engagement peak in 2025 reaching 1,683 hours, likely influenced by academic breaks or specific major releases during that period. While Simulation has historically been the participant's most-played category with over 824 hours, recent habits show a tactical shift toward the Shooter genre, which dominated the last 30 days of activity. Finally, the participant shows a clear preference for the PC (Windows) platform, which hosts the highest number of high-playtime outliers compared to handheld or console systems.

Table 6. Summary of Behavioral Interpretations and Statistical Key Findings

Aspect	Key Finding	Interpretation
Descriptive statistics	Mean (29.20 hrs, 35.42 sessions) much higher than median (3.81 hrs, 6 sessions).	A few games with very high engagement pull the average upward; most games receive minimal attention
Distribution shape	Right-skewed distributions for both variables.	Gaming engagement follows a power-law pattern; a small number of games dominate total playtime.
Temporal trends	Fluctuations in library growth and playtime across 2021–2026.	Gaming behavior may vary with academic schedule, game releases, or changing preferences
Correlation	Strong positive correlation ($\rho = 0.7942$, $p < 0.001$) between Play Count and Time Played.	Games that are enjoyed are played more frequently; repeated sessions accumulate into significant total playtime.
Platform variation	Differences in median playtime across platforms	May reflect game availability, platform-specific features, or personal preferences
Genre preferences	Categories differ in total engagement and change over time	Reflects personal interests, time availability, and possibly social factors

5.2 Comparison to Related Work

The findings from this personal gaming library analysis mirror broader digital consumption research, specifically the power-law distribution where a few items dominate total engagement. The strong positive correlation (ρ=0.7942) between Play Count and Time Played Hours aligns with established engagement patterns in digital media, confirming that session frequency is a reliable predictor of total time investment. Furthermore, temporal fluctuations in activity driven by academic schedules and content release cycles consistent with studies on student media habits, where external workload dictates consumption. Ultimately, these results demonstrate that personal gaming behavior follows predictable patterns like music, video, and other high-loyalty technology platforms.

5.3 Limitations

Several limitations should be acknowledged:

- Small Sample Size: The study is limited to a single player's library of 72 games, meaning the results represent personal habits rather than general population trends.
- Self-Report Bias: Although data was collected via Playnite, potential tracking errors or untracked activity may still affect the accuracy of the recorded engagement patterns.
- Missing Entries: The use of an "Uncategorized" label for games with missing metadata may have slightly influenced the precision of category-based findings.
- Short Data Collection Window: As a snapshot of 2021–2026, the analysis may not fully capture long-term longitudinal shifts in gaming preferences or habits.
- Platform-Specific Limitations: Variations in how different platforms integrate with Playnite could bias the results toward systems with better tracking support.
- Limited Variables: The research focuses on quantitative metrics like playtime but lacks qualitative context, such as game enjoyment or social influences.

5.4 Recommendations and Future Work

Suggestions for Other Students:

1. Expand Sample Size: Future studies should aggregate data from multiple individuals to improve generalizability and provide more reliable population-level insights.
2. Including Additional Variables: Incorporating metadata such as game ratings, completion status, and player reviews would provide a richer context for understanding engagement patterns.
3. Longitudinal Tracking: Observing gaming behavior over a span of 3 to 5 years would better distinguish temporary trends from lasting behavioral changes.

4. Cross-Platform Analysis: Systematically comparing behavior across different hardware could reveal how platform-specific features and availability influence play habits.

Alternative Methods for Future Research:

1. Machine Learning Approaches: Clustering algorithms can be applied to predict future game preferences and uncover hidden behavioral patterns not visible through standard statistics.
2. Social Network Analysis: Analyzing multi-user data can reveal how social connections and community trends affect game selection and total playtime.
3. Comparative Analysis: Investigating gaming patterns across different demographic groups or academic majors could show how personal characteristics influence digital consumption.
4. Qualitative Methods: Combining statistical data with interviews or surveys would provide deeper insights into the specific motivations and preferences behind observed habits.

6 Conclusion

This research successfully established a robust analytical framework for evaluating digital gaming behavior, effectively addressing the primary objective of modeling player engagement through descriptive and correlational statistics. The study characterized the library's distribution profile, proving that engagement is non-uniform and heavily concentrated on a select few high-engagement titles, such as Jurassic World Evolution 2 and Grand Theft Auto V. Statistical analysis revealed that while Simulation (824.2 hours) and Action (764.0 hours) are the dominant genres historically, a recent shift toward Shooter games (125.6 hours) highlights the evolving nature of the participant's interests over the analyzed period. By identifying these "outlier" titles and patterns, the study fulfilled the goal of distinguishing dominant games from the rest of the collection, reinforcing the stability and robustness of the quantitative findings.

Through this personal data analysis, the study observed that gaming habits follow a predictable power-law pattern where session frequency serves as a critical and reliable indicator of long-term engagement, supported by a strong Spearman Correlation ($\rho = 0.7942$). This insight reveals that time investment is not random but driven by specific "feedback loops" where repeated launches lead to deeper commitment. In daily life, these findings can be applied to improve time management by recognizing which genres like Simulation require significant long-term dedication versus those played in shorter bursts. Ultimately, this study demonstrates that applying non-parametric tools is an effective methodology for modeling digital engagement, providing a scientific baseline for understanding how technology use fluctuates alongside external factors like academic schedules.

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